Lecture 10

Instruments: the state after a reading

Objectives. *A POVM predicts a reading's probabilities but does not determine the state left for the next experiment. Lüders' update follows from a precise conditioning premise on all sub-events. An instrument specifies both statistics and state changes. We construct it from a pointer, compute the position pointer's added spread, and compare two full sequences with the same first POVM.*

- Prove Lüders' uniqueness from sub-event conditioning on a separable Hilbert space.
- Define an instrument and derive the instrument induced by a pointer model.
- Compute the Gaussian pointer's resolution and unread momentum spread, and the joint probabilities and final states of the qubit capstone.

10.1 Lüders as conditioning on every sub-event

QM-I, Postulate 4.1, postulated the ideal state update for a sharp reading. We now ask which probability-conditioning premise characterises it. Let ρ be a density operator on nonzero separable $\mathcal{H}$, and E a projection with $p = \mathrm{Tr}(\rho E) > 0$.

Theorem 10.1 (Lüders conditioning). *There is exactly one density operator τ satisfying*

$$\mathrm{Tr}(\tau E) = 1, \qquad \mathrm{Tr}(\tau E') = \frac{\mathrm{Tr}(\rho E')}{p} \quad \text{for every projection } E' \le E.$$

It is $\tau = E\rho E/p$. The first condition is also the case $E' = E$ of the second.

Proof. Expand $\rho = \sum_j r_j \left|\varphi_j\right\rangle\!\left\langle\varphi_j\right|$ by L1. The positive series $E\rho E = \sum_j r_j \left|E\varphi_j\right\rangle\!\left\langle E\varphi_j\right|$ has trace $\sum_j r_j \left\|E\varphi_j\right\|^2 = \mathrm{Tr}(\rho E) = p$, so its normalisation is a density operator. It has support in E. For $E' \le E$, pairing its series with E' gives the required conditional probability, since $EE' = E'E = E'$. Conversely expand a candidate $\tau = \sum_j t_j \left|\chi_j\right\rangle\!\left\langle\chi_j\right|$ in positive spectral terms. The identity $\mathrm{Tr}(\tau(\mathbb{1} - E)) = 0$ says that the sum of the nonnegative weights times $\left\|(\mathbb{1} - E)\chi_j\right\|^2$ vanishes. Every nonzero weighted χ_j lies in $\mathrm{ran}\, E$, hence $\tau = E\tau E$. For every unit $u \in \mathrm{ran}\, E$, use $E' = \left|u\right\rangle\!\left\langle u\right|$ to obtain $\langle u, \tau u\rangle = \langle u, \rho u\rangle / p$. Homogeneity extends equality of these quadratic forms to every vector in that subspace. Complex polarisation determines all their matrix elements there, and both candidate operators vanish on the perpendicular complement. Thus $\tau = E\rho E/p$. ∎

The premise concerns every subprojection, not just the elements of one chosen basis. Merely asking that the outcome E become certain permits many other states in its range when $\mathrm{rank}\, E \ge 2$. The theorem also does not assert that every real apparatus implements this conditioning rule, or explain a dynamical physical collapse. It characterises a rule once

the premise is imposed. At $p = 0$ it supplies no normalised conditional state. The proof works for a qubit and infinite-rank E; it uses no Gleason dimension restriction (compare BLPY pp. 211–212).

Self-check 10.1

1. Why does probability one for E force a density operator to have support in its range?
2. If $E = \mathbb{1}$, what does conditioning every sub-event require the output state to be?

10.2 An instrument carries probabilities and outputs

Definition 10.2 (Instrument)**.** For countable outcomes an *instrument* is a family of CP maps $\mathcal{I}_j$ on $\mathcal{T}_1(\mathcal{H})$ whose sum, convergent in every bounded expectation pairing, is a channel. For real Borel outcomes it is a family $B \mapsto \mathcal{I}(B)$ of CP maps, with total map $\mathcal{I}(\mathbb{R})$ a channel and countable additivity in every bounded pairing $\mathrm{Tr}(A\mathcal{I}(B)(\rho))$. For a density input, the outcome probability is $p(B) = \mathrm{Tr}\,\mathcal{I}(B)(\rho)$; if it is positive, the conditional state is $\mathcal{I}(B)(\rho)/p(B)$.

For explicit position computations we use the intervals and finite unions of L9. A full Borel position-pointer model uses the cited Borel spectral framework recalled in QM-II, §5.1; no measure-extension proof is claimed here. The general model argument below starts with a PVM already given. The sum or total map describes the *unread* experiment. Each outcome operation is generally trace decreasing on positive inputs, not a channel by itself. Normalising it makes the conditional rule nonlinear; its unnormalised CP map is what composes simply with the next operation.

The effect must depend on the operation, rather than on a chosen Kraus list. We reconstruct it from the operation's outcome trace. Here is why its probabilities form a POVM even without assuming a Kraus list. For an operation write $b(\varphi, \psi) = \mathrm{Tr}\,\mathcal{I}(B)(|\psi\rangle\langle\varphi|)$. Linearity gives a sesquilinear form, positive on equal arguments, with $0 \le b(\psi,\psi) \le \|\psi\|^2$, because the total instrument preserves trace and the complementary operation is positive. The quadratic-form Cauchy–Schwarz argument gives $|b(\varphi,\psi)| \le \|\varphi\|\,\|\psi\|$. As in L1, set $u_N = \sum_{n\le N} b(e_n,\psi)e_n$ in one ONB. Then $\|u_N\|^2 = b(u_N,\psi) \le \|u_N\|\,\|\psi\|$, so its coefficient series converges. Defining $F(B)\psi = \lim_N u_N$ yields a bounded linear operator with $\langle\varphi, F(B)\psi\rangle = b(\varphi,\psi)$ and $0 \le F(B) \le \mathbb{1}$. For a density's finite spectral head the pairing is $\mathrm{Tr}(\rho F(B))$; the remaining positive tail contributes at most its trace, proving the same equality for the full density. Instrument additivity and normalisation on rays now give the POVM axioms.

For a countable Kraus description the formula is simpler:

$$\mathcal{I}_j(\rho) = \sum_r K_{jr}\rho K_{jr}^\dagger, \qquad F_j = \sum_r K_{jr}^\dagger K_{jr}, \qquad \sum_{j,r} K_{jr}^\dagger K_{jr} = \mathbb{1} \tag{10.1}$$

with strong sums. L7 proves the total channel property; each positive outcome sum has the same trace-tail construction. The POVM retains the F_j while discarding the information about how each operation changes the state. That information matters for subsequent readings.

A *measurement model* specifies a meter space, one fixed ready density τ_M, a unitary U_M and a sharp pointer PVM $\Pi(B)$. Prepare $\rho \otimes \tau_M$ independently. The induced operation is

$$\mathcal{I}(B)(\rho) = \mathrm{Tr}_M\Big[(\mathbb{1}\otimes\Pi(B))U_M(\rho\otimes\tau_M)U_M^\dagger(\mathbb{1}\otimes\Pi(B))\Big]. \tag{10.2}$$

To prove CP explicitly, write $\tau_M = \sum_\ell s_\ell\, |m_\ell\rangle\langle m_\ell|$, choose a meter ONB (e_k) and set $P_B = \mathbb{1} \otimes \Pi(B)$, $J_\ell\psi = \psi \otimes m_\ell$. The coordinate operators

$$K_{\ell k}\psi = \sqrt{s_\ell}(\mathbb{1} \otimes \langle e_k|)P_B U_M J_\ell \psi$$

are bounded. Expanding product vectors in meter coordinates, as in L7, gives $\mathcal{I}(B)(\rho) = \sum_{\ell,k} K_{\ell k}\rho K_{\ell k}^\dagger$. Parseval gives their strong completeness sum $\sum_{\ell,k} K_{\ell k}^\dagger K_{\ell k} = \sum_\ell s_\ell J_\ell^\dagger U_M^\dagger P_B U_M J_\ell \le \mathbb{1}$; the latter convex sum converges in norm. L7's positive-series proof applies with total trace weight at most $\operatorname{Tr}\rho$, and with every finite spectator, proving CP. The partial trace of a positive output of trace below one means its trace times the partial trace of its normalisation (zero stays zero); complex-linear extension is well-defined by bounded local pairings. The total map is trace preserving. For disjoint pointer events, $A \otimes \mathbb{1}$ commutes with the pointer projections, so cross terms between their ranges vanish in every bounded local pairing. Strong convergence of orthogonal projection sums (norm convergence on each fixed vector) and the positive spectral-tail bound prove countable additivity. Thus this model induces an instrument, not merely a list of separate CP maps. A one-sided projection formula gives the same partial trace, but the sandwich makes the CP argument explicit.

Conversely every instrument on the countable or real Borel spaces used here has such a pointer model (*cited*: BLPY, Theorem 7.14 p. 160). BLPY, Theorem 10.1 p. 227, implements a POVM; the instrument theorem also fixes the output operations. Its general proof is not our finite Choi theorem. In finite dimension, adding an outcome register to L7's Kraus isometry gives the familiar construction. This establishes possible models, not uniqueness of an apparatus behind its statistics.

Self-check 10.2

1. Is $\mathcal{I}_j$ generally TP? Is its normalised conditional rule linear?
2. For a CNOT pointer, change the meter ready vector from $|0\rangle$ to $|1\rangle$ and derive the two outcome operations from (10.2).

10.3 The position pointer's instrument

Use L9's shear $\psi(x)\varphi(y - gx)$ in the impulsive limit, neglecting system and meter free evolution during the interaction. Take a normalised Schwartz meter φ and calibrated reading $z = y/g$, $g > 0$. The operation is

$$\mathcal{I}(I)(\rho) = \int_I K_z \rho K_z^\dagger \, dz, \qquad (K_z\psi)(x) = \sqrt{g}\, \varphi(g(z - x))\psi(x). \tag{10.3}$$

Each K_z is bounded since φ is bounded. The integral is first identified by its rank-one matrix-element pairings, which uniquely determine the trace-class output of the pointer model. Its bounded expectation pairings then follow for that operator. For a pure input, tracing the meter coordinates in gI gives the kernel

$$\psi(x)\overline{\psi(x')} \int_I g\varphi(g(z - x))\overline{\varphi(g(z - x'))}\, dz,$$

which is exactly the pairing of (10.3). Positive spectral sums and their probability tails extend it to every density operator. CP and additivity were already proved by the model, so no unintroduced operator-integration theorem is being assumed. The effect density is $K_z^\dagger K_z = g|\varphi(g(z - Q))|^2$, reproducing L9's reflected convolution after integration.

$K_z\rho K_z^\dagger$ is an operator *density* in the reading z; its trace is a probability density, not a dimensionless event probability. For a finite interval one integrates before normalising.

Where the density trace is positive, normalising it gives the usual posterior density rule for a resolved continuous readout; it is not division by the probability of the zero-width event $\{z\}$.

The same sharp Q PVM does not admit an exactly repeatable real-outcome instrument (*cited*: BLPY, Theorem 10.4 p. 248). This leaves legitimate Lüders operations for individual interval events. In particular, treating $I \mapsto E_Q(I)\rho E_Q(I)$ as one additive instrument fails: splitting an interval removes cross terms between its two subintervals. Approximate or finite-resolution repeatability is a different condition.

Self-check 10.3

1. What must be integrated before forming the conditional state of a finite interval?
2. Why is the family of separate interval Lüders operations not additive?

10.4 Resolution and unread momentum spread

In meter momentum coordinates the shear state is $\psi(x)\widetilde{\varphi}(p_M)e^{-igxp_M/\hbar}$, using the Fourier convention of QM-II. Ignoring the meter is independent of which meter basis is used for the partial trace. For a fixed p_M the system receives the phase $e^{-igp_MQ/\hbar}$, hence the momentum kick $-gp_M$. For a pure input the unread output momentum density is therefore

$$w_{\mathrm{out}}(p) = \int |\widetilde{\varphi}(p_M)|^2 w_{\mathrm{in}}(p + gp_M)\, dp_M = (w_{\mathrm{in}} * \kappa)(p), \qquad \kappa(q) = \frac{1}{g}\,|\widetilde{\varphi}(-q/g)|^2\,. \quad (10.4)$$

The Fourier shift rule verifies the sign; a positive mean p_M produces a negative system momentum shift. The same relation holds for density operators by their positive spectral sums. This is the unread channel; a *selected* reading multiplies the position amplitude by K_z and Fourier-transforms that product, rather than convolving its momentum density by κ.

For a centred Gaussian meter, K_z weights the position amplitude by a Gaussian centred at the indicated coordinate. The resulting momentum density depends on the phase and shape of that particular filtered amplitude. When the outcome is ignored, the meter momentum supplies the independent kick distribution instead. The two calculations answer different experimental questions even when they come from the same coupling.

Define the apparatus *resolution* as the standard deviation of its reading-error density, $\epsilon_Q = \Delta Q^{(2)}/g$, and its *added momentum spread* as that of the unread kick density, $\eta_P = g\Delta P^{(2)}$. Known means are calibrated or tracked separately. If the input momentum variance is finite, the ordinary variance rule for independent convolution gives $\mathrm{Var}_{\mathrm{out}}\, P = \mathrm{Var}_{\mathrm{in}}\, P + \eta_P^2$; the mean shifts by $-g\langle P^{(2)}\rangle$. These are statements about this model's reading error and added noise, not a replacement for the input state's own spreads.

For the centred unchirped Gaussian

$$\varphi(y) = (2\pi\sigma_M^2)^{-1/4} e^{-y^2/(4\sigma_M^2)},$$

$|\varphi|^2$ has variance σ_M^2, and its Fourier density has variance $\hbar^2/(4\sigma_M^2)$. The two densities in question are Gaussians, with

$$\epsilon_Q = \frac{\sigma_M}{g}, \qquad \eta_P = \frac{\hbar g}{2\sigma_M}, \qquad \epsilon_Q\eta_P = \frac{\hbar}{2}. \quad (10.5)$$

Sharper position resolution increases the unread added momentum spread. For any normalised Schwartz meter, QM-II's proved uncertainty relation (Theorem 4.9) gives $\epsilon_Q\eta_P \geq \hbar/2$.

More general finite-spread meters require its cited extension on $D(Q^{(2)}) \cap D(P^{(2)})$. Equality is not asserted for a chirped Gaussian. If one defines uncentred root mean square errors instead of standard deviations, biases also enter; they vanish in the centred Gaussian used for (10.5). This is a model-specific apparatus relation, distinct from a universal error–disturbance theorem and from the preparation inequality $\Delta Q \Delta P \geq \hbar/2$ for the system input.

Self-check 10.4

1. What happens to ϵ_Q and η_P when σ_M is halved at fixed g?
2. Does the convolution formula describe the selected or unread experiment?

10.5 Capstone: one first POVM, two sequences

Fix the input $\rho = |+x\rangle\langle+x|$, where $|+x\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. First read σ_z with a CNOT meter initially in $|0_M\rangle$: $|j\rangle\,|0_M\rangle \mapsto |j\rangle\,|j_M\rangle$, followed by the meter reading. The meter coordinate operators are $(\mathbb{1} \otimes \langle k_M|)(\mathbb{1} \otimes |j_M\rangle\langle j_M|)U_M(\psi \otimes |0_M\rangle) = \delta_{kj}E_j\psi$. Thus $\mathcal{I}_j(\rho) = E_j\rho E_j$, $E_j = |j\rangle\langle j|$, $j = 0, 1$. Each first outcome has probability $1/2$. Next rotate the system by

$$U = e^{-i\theta\sigma_y/2} = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \qquad \theta = \pi/3.$$

Finally read the noisy POVM $F_k = (\mathbb{1} + k\eta\sigma_z)/2$, $k = \pm 1$, with visibility $\eta = 1/2$ (distinct from added spread η_P) and specified operations $\mathcal{I}'_k(X) = M_k X M_k^\dagger$, where

$$M_+ = \mathrm{diag}(\sqrt{3}/2, 1/2), \qquad M_- = \mathrm{diag}(1/2, \sqrt{3}/2).$$

These positive square roots have $\sum_k M_k^\dagger M_k = \mathbb{1}$. The instrument, rather than just these effects, fixes the final states. The joint probabilities are

$$p(j,k) = \mathrm{Tr}\left(\mathcal{I}'_k(U\mathcal{I}_j(\rho)U^\dagger)\right) = \frac{1}{4}\left(1 + k(-1)^j\eta\cos\theta\right). \tag{10.6}$$

Indeed after the first reading and rotation the unit vectors are $\chi_0 = (\sqrt{3}\,|0\rangle + |1\rangle)/2$ and $\chi_1 = (-|0\rangle + \sqrt{3}\,|1\rangle)/2$. Applying M_k and normalising gives the complete table:

(j,k)	$p(j,k)$	final unit vector
$(0,+)$	$5/16$	$(3\,\|0\rangle + \|1\rangle)/\sqrt{10}$
$(0,-)$	$3/16$	$(\|0\rangle + \|1\rangle)/\sqrt{2}$
$(1,+)$	$3/16$	$(-\|0\rangle + \|1\rangle)/\sqrt{2}$
$(1,-)$	$5/16$	$(-\|0\rangle + 3\,\|1\rangle)/\sqrt{10}$

Each row's state is the normalised output of the *two* operations, not a state inferred from the second POVM alone. Summing either pair at fixed j gives $1/2$, and all four probabilities sum to one.

Now change only the first instrument to $\mathcal{J}_j(X) = K_j X K_j^\dagger$ with $K_j = |+x\rangle\langle j|$. Its effects are still $K_j^\dagger K_j = E_j$, so its first statistics agree for *every* input, not merely our chosen state. But both conditional outputs are $|+x\rangle$. After the same rotation their z mean is $-\sin\theta = -\sqrt{3}/2$, giving

$$p_{\mathrm{reset}}(j,k) = \frac{1}{4}\left(1 - k\frac{\sqrt{3}}{4}\right) = \frac{4 - k\sqrt{3}}{16}. \tag{10.7}$$

For either j, the final unit vector is

$$\frac{M_k U\,|{+}x\rangle}{\sqrt{(1-k\sqrt{3}/4)/2}}, \qquad U\,|{+}x\rangle = \frac{(\sqrt{3}-1)\,|0\rangle + (\sqrt{3}+1)\,|1\rangle}{2\sqrt{2}}.$$

The comparison separates the first outcome's probability from its conditional output. Figure 10.1 keeps the later rotation and second operation fixed. Its two routes represent different apparatus procedures, not two simultaneous branches of one experiment. Each first apparatus records the label j. Forgetting that label would require summing the unnormalised operations over j.

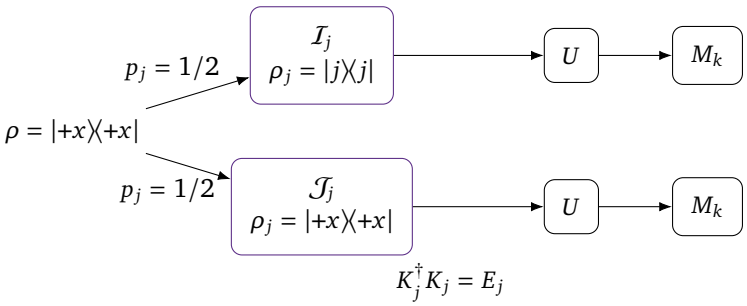


Figure 10.1: Two first instruments with effects E_j and the same first probabilities for the specified input. Their conditional outputs differ, so the common later operations U and M_k give different joint reading statistics.

These probabilities and states differ from the Lüders table even though the first POVM is identical. An observation's dynamical specification is its instrument. A second pure-ready meter can realise these operations by the isometry $\psi \mapsto \sum_k M_k\psi \otimes |k_M\rangle$. Retaining the two meter records coherently instead of conditioning them will be the comparison in L14.

A finite indirect measurement also permits a precise comparison between the input and output observables. Let A and B be Hermitian system matrices, let M be a Hermitian meter pointer, and let U_M be a unitary coupling. All expectations in this comparison use the initially prepared product state $\rho \otimes \tau_M$. Define the input operators and the output operators pulled back to that initial state by

$$A_{\text{in}} = A \otimes \mathbb{1}, \qquad B_{\text{in}} = B \otimes \mathbb{1},$$
$$M_{\text{out}} = U_M^\dagger(\mathbb{1} \otimes M)U_M, \qquad B_{\text{out}} = U_M^\dagger(B \otimes \mathbb{1})U_M.$$

The noise operator compares the indicated pointer with the input observable. The disturbance operator compares the final and initial versions of the other system observable:

$$N = M_{\text{out}} - A_{\text{in}}, \qquad D = B_{\text{out}} - B_{\text{in}}, \qquad \epsilon_A^2 = \langle N^2\rangle, \qquad \eta_B^2 = \langle D^2\rangle.$$

The roots satisfy $\epsilon_A, \eta_B \geq 0$. These root mean square quantities include any nonzero mean differences. They are defined on the joint prepared state, rather than inferred from the separate input and output distributions alone. They need not equal the standard deviations of the error and kick densities used for the Gaussian pointer.

Proposition 10.3 (Finite noise–disturbance bound). *For this finite matrix model, with input system spreads ΔA and ΔB,*

$$\epsilon_A \eta_B + \epsilon_A \Delta B + \Delta A \eta_B \geq \frac{1}{2} |\langle [A, B] \rangle_\rho|.$$

Proof. The final pointer and final system observable commute because the uncoupled operators act on different factors. Unitary conjugation therefore gives $[M_{\text{out}}, B_{\text{out}}] = 0$. Substituting the noise and disturbance definitions yields

$$[N, D] + [N, B_{\text{in}}] + [A_{\text{in}}, D] = -[A_{\text{in}}, B_{\text{in}}].$$

Take expectations, use the triangle inequality, and apply the finite-dimensional Robertson inequality to each commutator. The input product state's system spreads are ΔA and ΔB. Consequently

$$\frac{1}{2} |\langle [A, B] \rangle_\rho| \leq \Delta N \Delta D + \Delta N \Delta B + \Delta A \Delta D \leq \epsilon_A \eta_B + \epsilon_A \Delta B + \Delta A \eta_B.$$

The last inequality uses $\Delta N \leq \epsilon_A$ and $\Delta D \leq \eta_B$, since a variance is the mean square minus the squared mean. ∎

The mixed terms cannot generally be omitted. Sheet 10 computes a CNOT example with zero pointer noise and nonzero disturbance that distinguishes this bound from the naive product bound. This finite proof uses only Hermitian matrices; an unbounded-operator extension needs its own domain and moment hypotheses. The quantities and bound follow M. Ozawa, *Phys. Rev. A* **67**, 042105 (2003), arXiv:quant-ph/0207121.

The physics

Repeated ideal tests illustrate how an instrument matters to dynamics: QM-II's discrete Zeno computation uses its short-time survival law for $\psi \in D(H)$ and increasingly frequent selected projections. It is not an assertion that every physical continuous detector freezes the system. The optional extensions on Sheet 10 compute finite noise–disturbance quantities and the probability of a full sequence of selected projective tests.

Self-check 10.5

1. Verify the final vector for $(j, k) = (1, -)$ and its probability.
2. Which conditional outputs make the reset first instrument differ from Lüders, despite identical effects?

What is the state after a reading?

An observation is an instrument. Its POVM supplies probabilities, and its normalised positive-probability operations supply conditional states. Lüders' state is uniquely selected by conditioning every sharp sub-event. Every instrument has a pointer model, by the cited representation theorem. Two instruments with one POVM can give different later probabilities and states; resolution and unread disturbance also belong to the apparatus model rather than to a mean operator alone.

Summary. Sub-event conditioning proves Lüders' uniqueness without a dimension exception. Pointer projection sandwiches induce CP additive instruments. The Gaussian position

instrument has a continuous Kraus density; ignoring its record adds a reflected momentum-kick convolution. The centred unchirped pointer obeys $\epsilon_Q \eta_P = \hbar/2$. The capstone computes how a changed first instrument alters the rest of an otherwise fixed sequence.

Exercises. Try Sheet 10 in the separate *Exercise sheets* booklet (English).

Self-check answers.

10.1 1. Each nonnegative spectral contribution to the complementary probability must vanish. 2. The original ρ, since every projection probability is unchanged.

10.2 1. Generally no to both: the operation has outcome trace and its normalisation depends on the input. 2. The meter outcomes are reversed: $\mathcal{I}_j(\rho) = E_{1-j}\rho E_{1-j}$.

10.3 1. $K_z \rho K_z^\dagger$ over the interval. 2. The whole-interval sandwich has cross terms absent from the sum of the two smaller sandwiches.

10.4 1. Resolution halves, added momentum spread doubles. 2. Unread; selection multiplies an amplitude instead.

10.5 1. $M_- \chi_1 = (-|0\rangle + 3\,|1\rangle)/4$, norm squared $10/16$; multiply by first probability $1/2$ to get $5/16$. 2. Reset always leaves $|{+x}\rangle$; Lüders leaves the recorded $|j\rangle$.

Sources. Busch–Lahti–Pellonpää–Ylinen, *Quantum Measurement*, §7.7 pp. 159–161, Theorem 7.14 p. 160; conditioning pp. 211–212; position model pp. 241–243, equations (10.32)–(10.33) p. 242; repeatability Theorem 10.4 p. 248. Busch–Lahti–Mittelstaedt, pp. 41–43, 84–86. The direct density-operator conditioning proof and the fully specified qubit sequence are supplied here.